One sequence model for every HSP pull-down: what ESMC-6B predicts, and what it does not

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Summary

The task. One multi-task MLP on ESMC-6B protein embeddings (the 6-billion-parameter ESM3-family model) and its annotated sparse autoencoder (SAE). It predicts every non-drug quantity measured in the HSP pull-downs:

Kind	Targets	Runs
Baseline pull-down, no perturbation	pull-down level	GA_20, GA_22, GA_24, FS73, FS76 (HSPB1); GA_33 (DNAJA1, DNAJB11)
Abundance	lysate level, supernatant level	GA_22, GA_24, GA_33

Kind	Targets	Runs
Abundance-free binding	bound:free partition; enrichment over input	GA_22, GA_24 (partition); GA_33 (enrichment)
Effects	salt	GA_20, GA_22, GA_24
	temperature	GA_33
	Mg^{2+}	FS76

That makes 33 targets over 11,541 proteins. Drug effects are excluded.

Every number is out-of-fold. Folds are grouped by 30% sequence identity, so no homolog of a test protein is ever trained on. Each number is read against its own split-half ceiling.

Question	Answer
How much does sequence predict?	Mean R^2 / ceiling 0.27-0.28 over all 33 targets. Abundance 0.51; abundance-free binding 0.33; baseline pull-down 0.26; temperature 0.24; salt 0.23; Mg 0.17
Is it better than a linear read-out?	Yes, slightly: +0.028 [0.023, 0.034] over ridge on the best layer; most on salt (+0.039)
Does reading residue by residue help?	No. A learned attention-pooling front end over 7.4 M residue states equals the mean-pooled model: +0.000 [−0.003, +0.004]
Does one shared model help?	Marginally. Multi-task beats 33 single-task models on 22 of 32 targets, but by 0.003 R^2 on average
Do conditions share one representation?	Yes. A trunk trained without a run predicts that run nearly as well as one trained with it. The exceptions are the condition types no other run contains: heat (0.26 → 0.19) and Mg (0.15 → 0.10)
What predicts the pull-down with no perturbation?	Mostly how much protein there is. Measured abundance plus Tm beats sequence for every baseline. Sequence plus those measurements is best: R^2 0.30-0.54
Fold changes from sequence alone, no input information?	R2 0.12-0.27 per perturbation (rho 0.35-0.52), calibrated; top-10 % responders recovered 2.5-4x above chance; same accuracy as the full model (-0.002 [-0.007, +0.002])
What does sequence add beyond abundance?	Salt and Mg responses: +0.056 and +0.045 R^2 / ceiling over measured abundance and Tm alone
Regressing out the baseline	Salt and Mg responses are almost untouched (83-99% of their variance remains). Heat is not: the baseline accounts for about half of the 43 °C response and half of what sequence predicts about it
Is the plateau a ceiling on predictability?	No, a ceiling on *per-protein* information. The residual is strongly shared among members of the same complex (r 0.22-0.58, null ~0.00). One complex-mate term lifts mean R^2 / ceiling 0.309 → 0.410, far more than any architecture change
Does AlphaFold structure help?	No. Surface vs buried ion pairs, charged surface patches, Debye-Huckel screening at the real ionic strengths and Mg^{2+} carboxylate clusters all add 0.000 over the embedding; the largest residual correlation of any structural feature is 0.092
Does the native partner's interface help?	No. Measured on 29,816 experimental assemblies covering 6,306 proteins: how much exposed hydrophobic and aggregation-prone surface a native partner buries adds +0.002 (salt and heat: −0.000). Three pre-stated directional predictions all fail
Does a *predicted* chaperone-client interface help?	No, and the measurement is half noise. 4,787 Boltz-2 co-folds of the HSPB1 ACD dimer with tiled 16-mer client windows: all four pre-stated predictions fail, including real windows versus their own scrambles (-0.026, p = 0.955). The same window re-run moves by 0.140 ipTM, so half the window score is run-to-run noise
What the SAE features say	Pull-down and abundance: structured enzymes up, phosphorylated disordered Ser/Thr-rich regions down. Abundance-free binding, salt and Mg: membrane-helix features. Heat: structured α/β enzyme cores

The bottom line. Sequence predicts about a quarter of what these measurements reliably contain, and an MLP over 94 architectures, attention pooling and multi-task sharing all hit the same plateau. That plateau is not a limit on predictability, but the way past it is not more structure: AlphaFold surface electrostatics, ion pairs and mask-pooled residue states all add nothing, because the embedding already contains them. What the model misses is instead shared among the subunits of a complex, which no sequence model can reach because the folds separate homologs and complex-mates are not homologs. One complex-context term takes mean R^2 / ceiling from 0.31 to 0.41 for the quarter of proteins in a curated complex, though it fades to +0.007 across the broad STRING physical network. That complex effect is not an interface effect: on 29,816 experimental structures, how much surface a native partner buries explains none of the residual and only +0.014 of the complex-shared part, so what a complex contributes is something other than steric competition for the chaperone's binding surface. Predicting the missing structure does not rescue it either: 4,787 co-folds of the HSPB1 ACD dimer with tiled client windows fail every pre-stated test, and cannot tell a real window from its own scramble, so three independent structural approaches now land in the same narrow band just above nothing (largest residual correlation 0.092, 0.087 and 0.134). The no-perturbation pull-down is mostly abundance and stability. The salt and Mg responses are where sequence carries information that the measured lysate properties do not.

1. Data, targets and features

Proteins. The union of every run is 11,570 proteins, of which 11,541 have a sequence and all features. Embeddings and SAE features already existed for 9,879. I computed the rest in one pass of ESMC-6B (1,662 proteins, under 1 minute on one RTX card).

A control re-embedding of 60 stored proteins reproduced them: median r 1.0000. One protein moved slightly with batch context (r 0.98). Embedded alone it sits between its two batched versions, so this is bf16 numerics, not a pipeline error.

Targets (`cm02`). 33 per-protein quantities, each with a split-half reliability across proteins (the ceiling):

Kind	n	Ceiling range
Abundance (GA_33 input, GA_22/24 supernatant)	3	0.98-0.995
Baseline pull-down (no perturbation)	7	0.98-0.998
Abundance-free binding (GA_22/24 partition, GA_33 enrichment)	4	0.99
Salt (GA_20 75/150 mM, GA_22/24 60/120 mM in pull-down, supernatant and partition)	14	0.76-0.99
Temperature (GA_33, 37 and 43 vs 35 °C, both baits)	4	0.64-0.98
Mg^{2+} (FS76)	1	0.98

Drug arms were excluded throughout, with one exception: FS76's Mg effect is averaged over the lithium doses, because lithium has no proteome-wide effect (reliability 0.14). GA_22 and GA_24 effects use only their drug-free arms and carry the drift correction of the GA_22/24 report.

Inputs.

- ESMC-6B hidden states 0, 10, . . . , 80, mean-pooled.
- The layer-60 SAE: 16,384 features, max over residues.
- Per-residue final-layer states for the attention model: 7.36 M residues, 37 GB.
- Sequence covariates: length, charge, pI, hydropathy, composition.
- Optionally, measured per-protein properties: lysate abundance here, PaxDb abundance, Meltome Tm.

2. The model and how it was trained

One MLP:

- an input branch per layer (a learned projection) and for the SAE;
- a shared trunk ending in a 64-dimensional latent;
- one linear head per target, reading that shared latent.

The loss is masked MSE over all observed targets, each standardised on the training fold.

Efficient training. Each architecture is trained as a population of 72-144 members, differing in learning rate, weight decay, dropout and seed. All members are one batched computation: every weight carries a member axis, and a vectorised AdamW applies per-member hyperparameters.

All inputs sit on the GPU. Each member is early-stopped on an inner validation fold, and the outer test fold is predicted by the mean of the eight best members (chosen on validation only).

GPU choice. RTX PRO 6000 (96 GB) for everything:

- data plus a population is 2-50 GB;
- the attention model holds 37 GB of residue states plus 15-25 GB of work;
- nothing needed an H100 or H200.

Job type	SM active	Occupancy	Tensor active	DRAM active	Peak memory
MLP populations	0.84-0.95	0.49-0.58	0.04-0.05	0.76-0.87	3-50 GB
Attention MLP	0.95	0.48	0.21	0.66	54-66 GB
ESMC-6B embedding					18.5 GB at 12-17 k residues/s

The populations are memory-bandwidth-bound: the matrices per member are small. Packing more members per step was the lever, and it kept SM activity near the maximum.

Scale. About 10 GPU-hours on at most 8 RTX cards, in roughly three hours of wall clock:

- 94 MLP architectures;
- 7 attention configurations;
- 33 single-task models;
- 6 leave-one-run-out trunks;
- 6 residualisation and covariate models.

I stopped four sweep jobs early. Their SAE-input groups cost 10-20 minutes each and had already shown that the SAE adds nothing to the dense layers.

3. How much sequence predicts

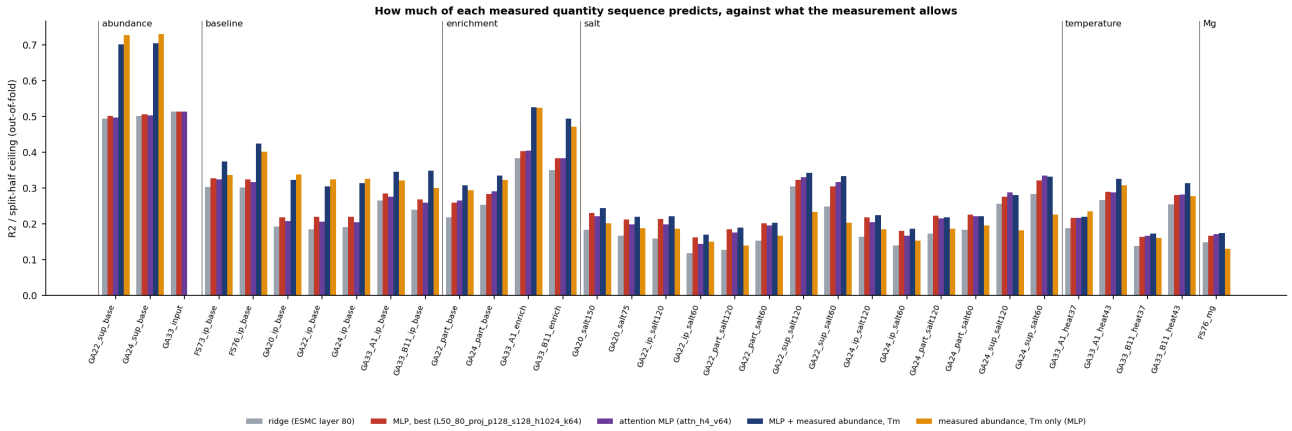


Figure 1: Out-of-fold R^2 / ceiling for every target: ridge on ESMC layer 80; the best MLP; the attention-pooling MLP; the MLP with measured abundance and Tm added; measured abundance and Tm alone.

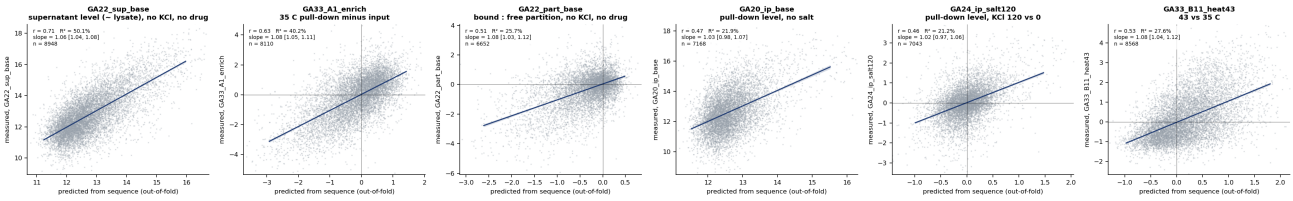


Figure 2: Predicted (out-of-fold) against measured for one target of each kind, with the least-squares line and 95% band.

Mean R^2 / ceiling, all targets, with paired cluster-bootstrap intervals (1,000 resamples of sequence clusters):

Model	All	Abund.	Base.	Enrich.	Salt	Temp.	Mg
Ridge, ESMC layer 80	0.245	0.504	0.240	0.302	0.191	0.213	0.149
MLP, layers 50+80	0.273	0.506	0.261	0.333	0.229	0.238	0.168
Attention MLP	0.273	0.505	0.257	0.336	0.230	0.239	0.172
Abundance + Tm only (32 t.)	0.286	0.729	0.336	0.403	0.186	0.246	0.131
Sequence + abundance + Tm (32 t.)	0.316	0.703	0.348	0.416	0.243	0.258	0.176
Sequence only (32 t.)	0.263	0.497	0.257	0.329	0.228	0.233	0.167

The models marked "32 t." are trained on 32 targets; the MLP row is the multi-task model with 144 members; the lysate abundance target is dropped because it is the abundance input.

Comparison	Difference in R^2 / ceiling, 95% interval
MLP − ridge	all +0.028 [0.023, 0.034]; salt +0.039 [0.028, 0.050]; enrichment +0.031; abundance +0.002 (n.s.)
Attention − mean-pooled MLP	all +0.000 [−0.003, +0.004]; no kind differs

Comparison	Difference in R^2 / ceiling, 95% interval
Sequence + measurements − measurements alone	salt +0.056 [0.042, 0.072]; Mg +0.045 [0.024, 0.065]; baseline +0.012 (n.s.); abundance −0.026
Measurements − sequence	abundance +0.233; baseline +0.079; enrichment +0.074; salt −0.042; Mg −0.036

A plateau. All 94 MLP architectures lie within 0.274-0.276 of the best. That holds across:

- layer sets (50 + 80 or all nine);
- projection widths of 32-256;
- latents of 16-256;
- trunks of 256-2,048;
- 150 or 300 epochs.

The SAE as an input is worse than the dense layers: 0.18 alone, nothing added on top. Attention pooling over residues, at 4, 8 or 16 heads (seven configurations, 0.268-0.273 of ceiling), equals mean pooling. The ensemble's members are nearly interchangeable (best member vs top-8 mean: r 0.9996).

The limit is what ESMC-6B's representation carries about these measurements, not the read-out.

The one weakness. Adding sequence to the measured covariates lowers abundance (0.729 → 0.703) and a few individual baselines. The combined model's regularisation does not fully adapt when one input block is far more informative than the other. A residual design (sequence predicting what the covariates leave) would avoid this.

3b. Fold changes alone: each perturbation from sequence, with no input information

This model knows nothing about the input / lysate distribution. One MLP is trained only on the 11 pull-down fold changes, with no abundance, supernatant, partition or baseline targets and no measured covariates: ESMC-6B layers 50 + 80 and sequence covariates in, 11 fold changes out (`cm11`, run `s5`).

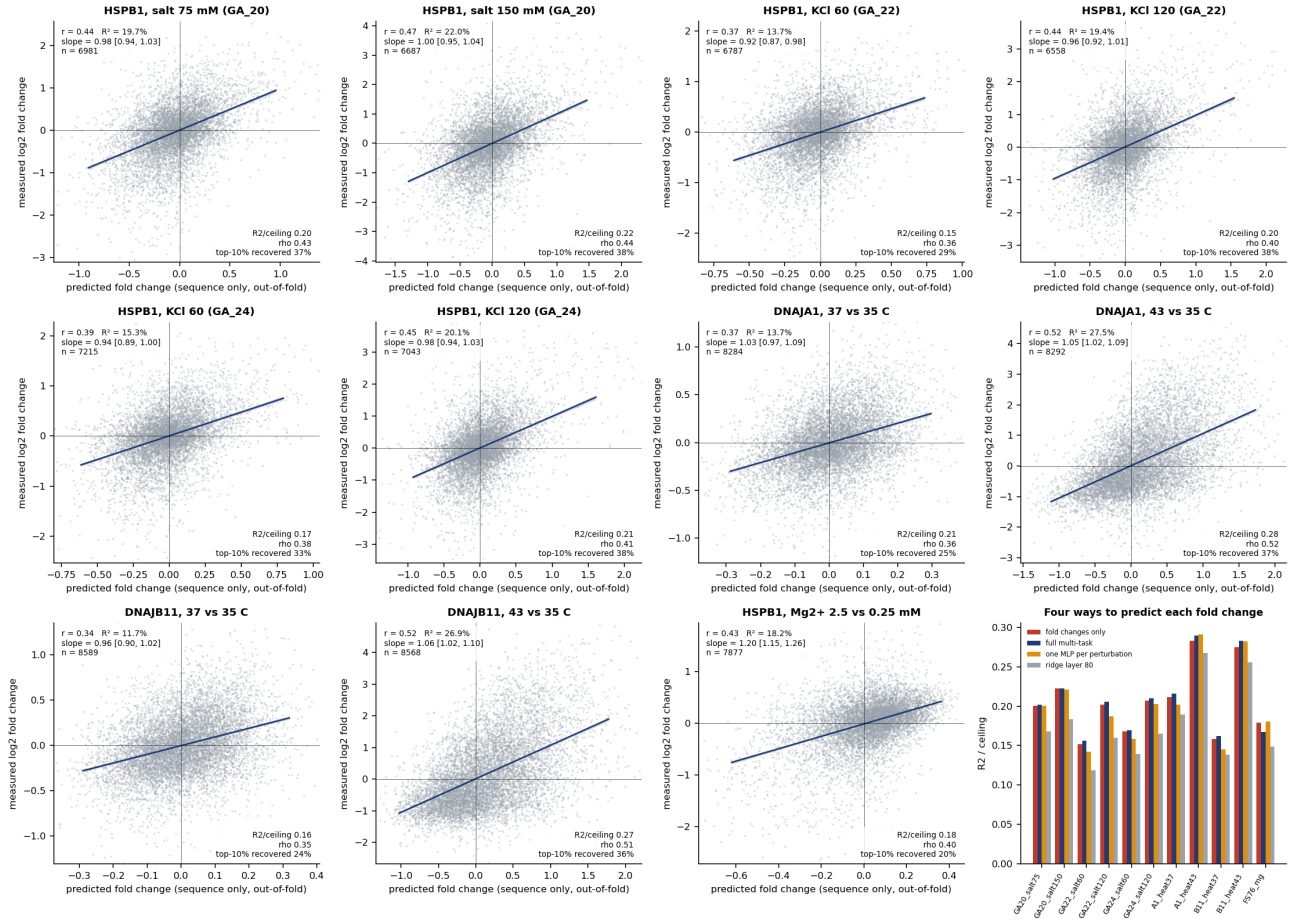


Figure 3: Predicted against measured log2 fold change for every perturbation (sequence only, out-of-fold), with the least-squares line; last panel: the same targets from four models.

Perturbation	Ceiling	R2	R2/ceil.	rho	Top 10 %	Bottom 10 %	Sign
HSPB1, salt 75 mM (GA_20)	0.98	0.197	0.20	0.43	37 %	30 %	79 %
HSPB1, salt 150 mM (GA_20)	0.99	0.220	0.22	0.44	38 %	29 %	80 %
HSPB1, KCl 60 (GA_22)	0.90	0.136	0.15	0.36	29 %	27 %	76 %
HSPB1, KCl 120 (GA_22)	0.96	0.193	0.20	0.40	38 %	27 %	78 %
HSPB1, KCl 60 (GA_24)	0.91	0.152	0.17	0.38	33 %	27 %	76 %
HSPB1, KCl 120 (GA_24)	0.97	0.201	0.21	0.41	38 %	25 %	79 %
DNAJA1, 37 vs 35 C	0.64	0.136	0.21	0.36	25 %	28 %	79 %
DNAJA1, 43 vs 35 C	0.97	0.274	0.28	0.52	37 %	33 %	89 %
DNAJB11, 37 vs 35 C	0.74	0.117	0.16	0.35	24 %	25 %	77 %
DNAJB11, 43 vs 35 C	0.98	0.268	0.27	0.51	36 %	27 %	89 %
HSPB1, Mg2+ 2.5 vs 0.25 mM	0.98	0.176	0.18	0.40	20 %	37 %	77 %

“Top / Bottom 10 %”: of the proteins measured in the top (or bottom) 10 %, the share the prediction also places in its top (bottom) 10 %; chance is 10 %. “Sign”: among the 10 % of proteins with the largest measured change, the share whose predicted change is on the same side of the median.

- **Sequence alone predicts every fold change, modestly**: R2 0.12-0.27 (0.15-0.28 of ceiling), rank correlation 0.35-0.52. The predictions are calibrated (out-of-fold slopes 0.92-1.06; Mg 1.20), so the size of a

predicted change means what it says.

- **As a screen it is 2.5-4 times better than chance**: a quarter to over a third of the strongest responders in each direction land in the predicted top or bottom 10 %, and the direction is right for 76-89 % of the strongest. Heat at 43 C is the most predictable perturbation; KCl 60 mM and heat 37 C, the smallest effects, the least. Mg is asymmetric: proteins that fall with Mg are found far better (37 %) than those that rise (20 %).
- **Leaving out the input information costs nothing**: the fold-change-only model equals the full multi-task model (difference in mean R2 / ceiling -0.002 [-0.007, +0.002]), matches one MLP per perturbation (+0.004 [-0.000, +0.008]) and beats ridge (+0.030 [0.022, 0.037]). The abundance and baseline targets did not help the network learn the fold changes, consistent with section 5.3: salt and Mg responses are largely independent of the baseline.

3c. What the model misses is largely a property of the complex, not the protein

Section 3 reports a plateau: 94 architectures, attention pooling over every residue, single- against multi-task, all land within 0.274-0.276 of ceiling. A plateau can mean the representation is exhausted, or that the remaining variance is not a function of the protein's sequence at all. The second possibility is testable (`cm12`).

A pull-down measures a protein that may be held in a multi-subunit assembly, so its level, and its response to a perturbation, can be a property of the assembly. Sequence cannot reach that. The design makes the test clean: the outer folds are grouped by MMseqs2 30 % identity and the subunits of a complex are unrelated in sequence, so complex-mates scatter across folds and no complex-level effect is learnable from sequence by construction. Any such effect therefore lands entirely in the residual.

For every target, each protein's residual (measured minus out-of-fold prediction) is correlated with the mean residual of its other Complex Portal complex-mates, taken **within the same outer fold** so that both sides are held out of the same model. The null is not zero – residuals share abundance and other structure – so it is obtained by shuffling residuals within fold, 500 draws, which preserves every complex's size and each protein's fold while destroying the pairing.

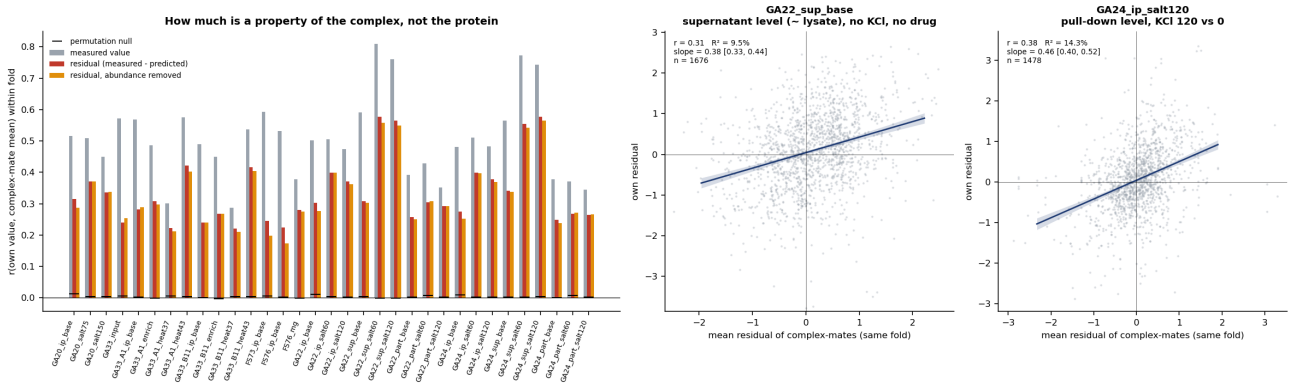


Figure 4: Left: how much of the measured value, and of the residual, is shared with complex-mates, against the permutation null (black dashes). Right: two examples.

Target kind	n	Residual r	Null	R2/ceil.	+ complex	Gain
baseline	7	0.27	0.007	0.384	0.440	+0.055
abundance	3	0.30	0.004	0.557	0.584	+0.027
enrichment	4	0.27	-0.001	0.250	0.299	+0.049
salt	14	0.40	0.003	0.281	0.445	+0.164

Target kind	n	Residual r	Null	R2/ceil.	+ complex	Gain
temperature	4	0.32	0.004	0.193	0.266	+0.073
Mg	1	0.28	-0.001	0.113	0.213	+0.100
all 33	**33**	**0.34**	**0.004**	**0.309**	**0.410**	**+0.101**

Scored on the 2,858 proteins that are in a Complex Portal complex with at least two measured members; "+ complex" adds one cross-fitted term, the complex-mate mean residual, with its coefficient fitted on the other outer folds.

- **The residual is strongly complex-structured for every target**: r 0.22-0.58 against a null of ~0.00, z = 6.3-16.9. The effect is largest where the sequence model is weakest.
- **It is worth more than everything architecture gave**: mean R2 / ceiling rises 0.309 to 0.410. On the salt responses it roughly doubles R2 (GA_22 KCl 60: 0.19 to 0.37; GA_24 KCl 60: 0.17 to 0.35), and on heat at 43 C it rises from 0.22-0.24 to 0.31-0.35. For comparison, the entire MLP-over-ridge gain is +0.028.
- **It is not abundance.** Projecting out `abundance_here`, PaxDb abundance and length barely moves it (mean r 0.336 to 0.326).
- **It is not shared peptides or paralogues.** The deployable variant, in which a protein's partner mean is taken only from **training-fold** proteins – measurements already in hand, never in the test protein's own 30 % identity cluster – performs identically (0.410 against 0.403). That version is also the honest one: it uses no held-out measurement.

Two limits. This is measured on the 25 % of proteins Complex Portal annotates with a measured partner; Complex Portal is incomplete, so the coverage is a floor, not a statement about the other 75 %. And the term consumes measured partner data, so it improves annotation of a *measured* proteome, not the sequence-only prediction of an unmeasured one – for a complex with no measured member it offers nothing.

How far does it reach? Complex Portal annotates only a quarter of the measured proteins, so the test was repeated on the STRING v12 *physical* subnetwork at three confidence cutoffs, which buys coverage at the cost of specificity (`cm16`, same training-fold-only design).

Graph	Proteins	Median degree	Residual r	R2/ceil.	+ partner	Gain
Complex Portal	2,858	5	0.346	0.300	0.401	+0.101
STRING physical >= 900	6,014	4	0.187	0.291	0.323	+0.032
STRING physical >= 700	7,341	6	0.145	0.286	0.306	+0.019
STRING physical >= 400	10,165	12	0.085	0.280	0.287	+0.007

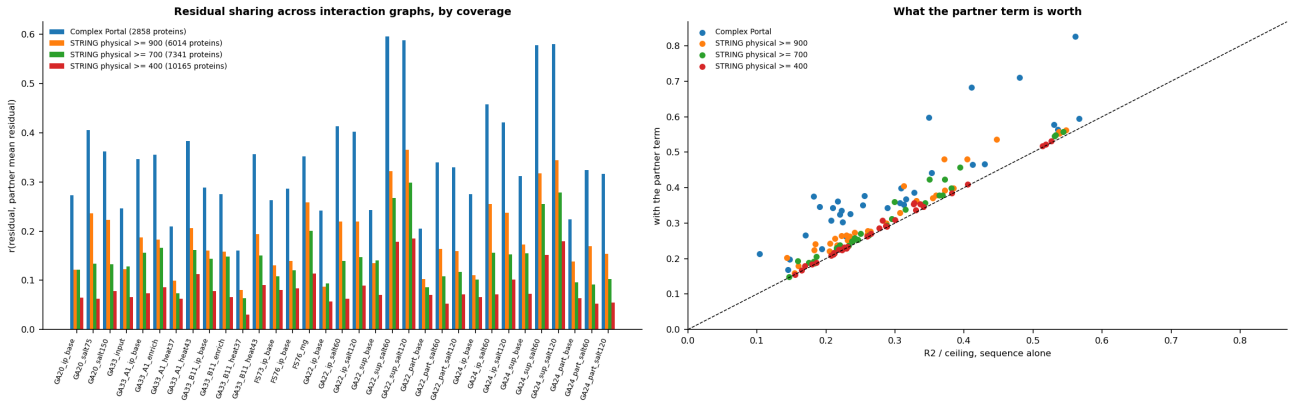


Figure 5: Residual sharing by graph. Left: per target, the four graphs ordered by coverage. Right: R2 / ceiling with the partner term against sequence alone; points above the diagonal are gains.

The effect is **specific to curated stable complexes and decays fast as the graph widens**. STRING reaches 10,165 of 11,570 proteins, but at that coverage a protein's partner mean is an average over a dozen loose associations and is worth +0.007. This tempers the claim above: complex context is a large lever for the quarter of the proteome that sits in a defined assembly, and a small one for the rest. It is not a general route to the other three quarters, and the honest summary of the campaign-wide number is that 0.41 applies to 2,858 proteins.

The conclusion for the campaign is that the ~0.27 plateau is the ceiling on *per-protein* sequence information, not on predictability. Interaction context is a larger lever than any change to the network.

3d. Structure, tested directly: it adds nothing

A simple probe motivates this section. Of the sequence covariates, net charge per residue is the single strongest predictor of the salt response, reproducibly across three independent experiments (Spearman rho 0.20-0.23 for GA_20, GA_22 and GA_24). But the *capacity* to form ion pairs is not: 2 x min(D+E, K+R) gives rho -0.04 to -0.10, and the total charged fraction -0.02 to -0.09. And every one of these is orthogonal to the sequence model's residual (|rho| <= 0.07), so the network has already absorbed the compositional charge signal completely. For salt its prediction is close to a pure charge axis: rho(prediction, measured) 0.36-0.44 against rho(prediction, net charge) 0.38-0.41.

Whatever is left, if it is a protein property at all, therefore has to be spatial – *where* the charge sits, whether an ion pair is buried or solvent-facing, how large a contiguous charged patch is. None of that survives averaging over residues. So the AlphaFold human proteome was folded in (`cm13`): 20,256 models parsed, none failed, and the model sequence is byte-identical to the embedded sequence for 20,001 of them, so per-residue quantities cannot be silently misaligned. Features per protein: relative SASA and burial; ion pairs split into buried and surface; surface charge, buried charge, the most charged 10 A surface neighbourhood, dipole; the change in intramolecular Debye-Huckel energy between the two ionic strengths **actually used in each experiment** (kappa = 0.329 sqrt(I) /A, with MgCl2 contributing I = 3c); carboxylate clusters as the geometric signature of a Mg^{2+} site; exposed and buried hydrophobic patches; pLDDT disorder and domain counts; crude secondary structure.

The test is not whether these predict the response – they do, because they correlate with composition – but whether they predict the **residual** of the out-of-fold sequence prediction (`cm14`, cross-fitted ridge, alpha chosen inside the training folds).

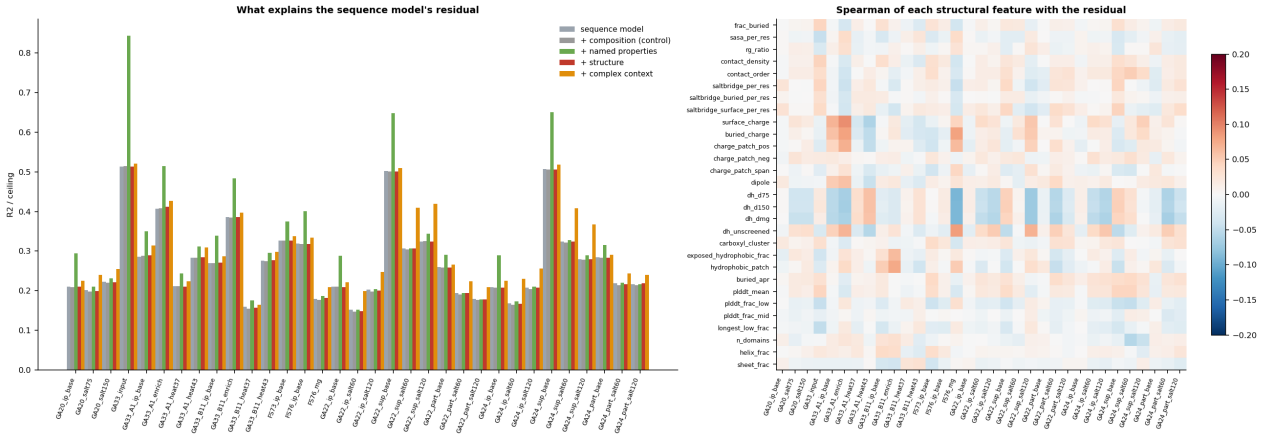


Figure 6: Left: R2 / ceiling for each target with each block added to the sequence model. Right: Spearman of every structural feature with the residual, on a +-0.2 scale.

Added to the sequence model	Mean R2 / ceiling, 33 targets
nothing (the sequence model)	0.273

Added to the sequence model	Mean R2 / ceiling, 33 targets
composition, 25 covariates (negative control)	0.271
AlphaFold structure, 28 features	**0.272**
named properties (abundance, Tm, keywords)	0.320
complex context	0.304
structure + named	0.322
everything	0.348

- **Structure adds nothing.** The bootstrap interval straddles zero on almost every target, and structure over the named properties is likewise null. The composition control behaves identically, which is the evidence that the test can detect "already absorbed" rather than being insensitive.
- **Resolving ion pairs in 3D does not rescue the hypothesis.** Averaged over the 14 salt targets, every charge and ion-pair feature correlates with the measured response and with nothing in the residual:

Feature	rho with measured	rho with residual
surface charge	+0.167	+0.027
buried charge	+0.152	+0.016
largest positive surface patch	+0.075	+0.010
salt bridges per residue	-0.041	+0.004
salt bridges, buried only	-0.006	+0.003
salt bridges, surface only	-0.048	+0.006
Debye-Huckel dE at 150 mM	-0.080	-0.035
carboxylate clusters (Mg^{2+})	-0.042	+0.012

The largest residual correlation of **any** structural feature with **any** target is 0.092.

Pooling the embedding over structural masks does not help either. Instead of averaging the ESMC states over the whole protein, they were pooled separately over surface, buried, ordered, disordered, charged-surface, hydrophobic-surface and buried-hydrophobic residues (`cm15`), giving the network the masks rather than making it learn them. On the 11 fold changes: sequence alone 0.2053, + structural features 0.2068, + mask pooling 0.2078, + both 0.2066. The internal check that the "all" mask reproduces the stored mean-pooled layer 80 gives r = 1.0000.

The caveat that matters. An AlphaFold model is itself predicted from sequence, and protein language models are known to encode structure implicitly, so this is close to asking whether one sequence-derived view adds to another. The result says that nothing these summary statistics extract from a predicted **single-chain, unbound, unmodified** structure is missing from the embedding. It does not say 3D geometry is irrelevant to salt sensitivity. The quantity the salt-bridge hypothesis is really about is the electrostatics of the chaperone-client *interface*, which needs a structure of the complex, not of the client alone.

3e. The native partner, tested on experimental structures: it adds nothing either

Section 3d left one escape route open: an AlphaFold model is predicted from the same sequence the network reads, so a null there is half-expected. A **complex** is not. It is a function of two sequences and their pairing, and the pairing is outside the model's input entirely – which is exactly why the complex-mate term in 3c worked. So the next question is whether the missing variance is an *interface* property.

The hypothesis is competition. HSPB1 binds exposed hydrophobic and aggregation-prone surface on non-native clients. A protein that spends its life with that surface buried against a native partner should be a poor client, and a perturbation that loosens the native interface should make it a better one. That predicts a specific thing about the residual, and it is testable with structures that already exist – no prediction, no GPU, 17 minutes of CPU.

What was measured (`cm17`, `cm18`). Every PDB biological assembly containing at least one measured protein, capped at 15 structures per protein, is 29,816 assemblies. For each measured chain the solvent-accessible surface was computed twice – alone, and in the presence of its partners – and the difference resolved by residue class: total buried area, the fraction of *exposed hydrophobic* surface a partner occludes, the same for aggregation-prone regions, the composition and ion-pair density of the interface itself, and occlusion by nucleic acid. That is 15 features, each aggregated over structures as median, max and min, for **6,306 of the 11,570 proteins** (1,816 of which have no interface in any structure – the informative zero, deliberately kept).

Entries holding at least *one* measured protein, not two: requiring a measured partner would condition on having a partner, which is close to the variable under test. Whether a partner chain was itself measured is irrelevant to how much surface it buries.

The guards. Chains were aligned to their UniProt sequence (median identity 1.000, p05 0.987; 546 of 110,832 chains fell below 0.80 and were dropped). RCSB names symmetry copies `A-2` where SIFTS calls them `A`; matching on the raw name would have silently discarded exactly the symmetry-generated partners that matter most, such as a homodimer absent from the asymmetric unit. The 10,462 chains with no partner chain in their assembly come out with a mean interface fraction of 0.00008, as they must.

The positive control. Before asking whether the features explain anything, they have to agree with things already known. Complex Portal members bury 0.176 of their surface against 0.090 for non-members, and burial correlates with Meltome T_m at rho +0.170 (n = 5,407). The measurement is real.

The result is a null, and a flat one (`cm19`, mean R^2 / ceiling over 33 targets):

Block added to the sequence model	R^2 / ceiling
sequence alone	0.278
+ study-depth control (structure count, chain size, partner count)	0.278
+ composition control	0.278
+ native-partner interface	**0.280**
+ named properties (abundance, T_m, keywords)	0.326
+ complex-mate term (3c)	0.319
+ everything	0.353

By kind, the interface block adds +0.002 to baselines, +0.001 to enrichment, +0.007 to abundance, and **−0.000 to both salt and temperature** – the two response types the hypothesis was built for. Every per-target bootstrap interval for the interface gain over the depth control includes zero.

The three directional predictions, fixed before looking, all fail.

Prediction	Feature	Targets	rho with residual
D1 salt releases an ionically held interface	interface ion pairs per 1000 Å^2	14 salt	−0.009 to +0.025, every CI includes 0
D2 heat unmasks occluded aggregation-prone surface	fraction of APR surface occluded	4 temperature	−0.022 to +0.008
D3 leftover hydrophobic surface is what HSPB1 binds	exposed hydrophobic area in the bound state	7 baselines	−0.011 to +0.014

D3 is worth a second look because it fails in an interesting way: on the *measured* scale it correlates **negatively** with the HSPB1 baselines (−0.050 to −0.081 for FS73, FS76, GA_20, GA_22, GA_24), the opposite of the prediction. Proteins with more hydrophobic surface left exposed in their native complexes are pulled down slightly *less*. That sign is consistent with those proteins being small, abundant and stable rather than with any chaperone-competition story, and it vanishes entirely in the residual (+0.009), so the sequence model already accounts for it.

The largest residual correlation of **any** of the 46 interface features with **any** of the 33 targets is 0.087 – essentially the same ceiling as the 0.092 found for AlphaFold monomer features in 3d.

One near-miss, and why it is not a result. Interface features predict the complex-mate mean residual of 3c at r = +0.080. But a control block built only from study depth and chain size reaches +0.066, so the interface contributes +0.014. In an earlier version of this analysis that gap looked like +0.04; the difference was an artefact of aggregating over however many structures a protein happens to have, since a maximum over 311 structures is mechanically larger than one over 3. Subsampling to a fixed cap before aggregating removed most of the apparent effect. The honest reading is that the complex-level signal of 3c is **not** explained by how much surface the partner buries.

What this does and does not close. It closes the version of the competition hypothesis that can be answered with static, unbound-versus-bound surface burial in native complexes, measured on more than half the proteome. It does not touch the chaperone side: none of these structures contains HSPB1, and the quantity the hypothesis is ultimately about is the interface between the chaperone and a *non-native* client – a state that is essentially absent from the PDB, because it is transient, low-affinity and disordered. Testing that needs a model of the bound complex, which is the next tier and a genuinely different experiment.

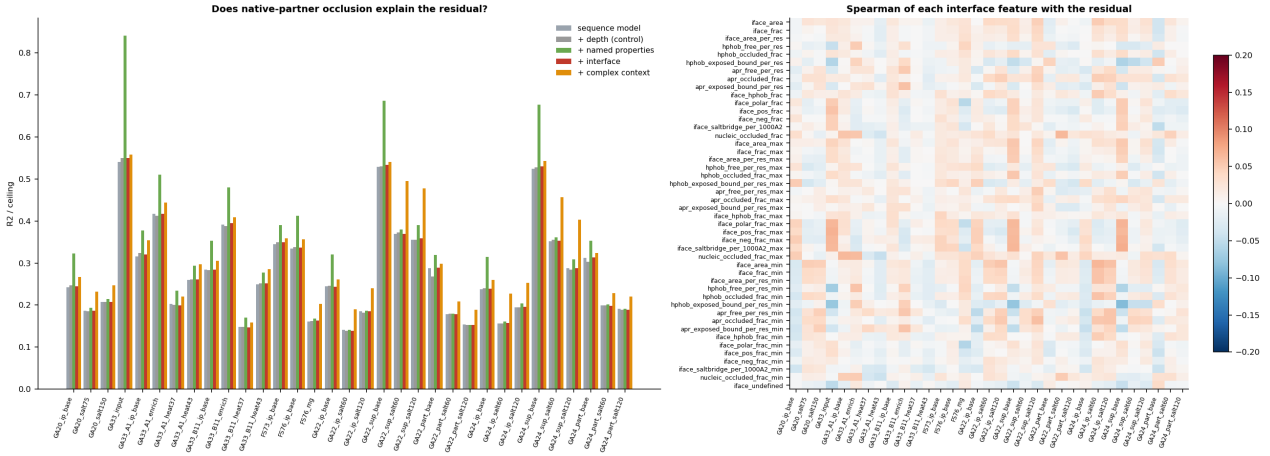


Figure 7: Native-partner occlusion against the sequence model's residual. Left: R^2 / ceiling per target; the interface bars sit on the sequence-only bars, while named properties and the complex-mate term rise above them. Right: Spearman correlation of each of the 46 interface features with the residual, on the same colour scale as the structure heat map in 3d.

3f. The chaperone's own interface, predicted: a null, and a measurement too noisy to rest on

3e closed the version of the competition hypothesis that experimental structures can answer, but it left the question the hypothesis is really about untouched: the interface between HSPB1 and a *non-native* client. No such structure exists in the PDB. This tier predicted it (`cm20-cm25`).

Design A. The HSPB1 alpha-crystallin domain (ACD, residues 84-171) as a dimer – the client-binding unit, and exactly the construct crystallised in 9ef3 and 3q9p – co-folded with one short window of a client, using Boltz-2 in single-sequence mode. A short window is the point: HSPB1 binds locally unfolded segments, so a full-length co-fold would dock the chaperone onto a *folded* client surface, which is the state 3e already showed is the wrong one. 200 proteins, 20 from each decile of the measured bound:free partition, every sequence tiled into non-overlapping 16-mers, 4,787 co-folds in 3.98 h on one RTX PRO 6000 at 97.7 % mean utilisation and 13.0 GB of 96 GB. No job failed.

Windows were tiled **uniformly**, not chosen by hydrophobicity or aggregation propensity. Choosing them would have confounded the structural score with the composition that selected them, so that the score would "predict" binding for the same reason a hydrophobicity scale does – which the sequence model already contains.

The readout is the bound:free partition of GA_22 and GA_24, because the partition divides abundance out experimentally rather than statistically: a protein scores high because it binds HSPB1, not because there is a lot of it. The two runs are independent measurements of the same quantity and agree at r = 0.914, which is the ceiling this tier could reach.

The validation gate, passed before any GPU time was committed to the pilot

9ef3 is the HSPB1 ACD with its own C-terminal IxI motif peptide bound in the beta4/beta8 groove. It was **released 2026-06-24, after Boltz-2 was trained**, so it is a genuine held-out test rather than a recall exercise. The groove was not asserted from the literature; it was measured as the 20 residues of 9ef3 chain A within 4.5 A of the peptide.

Check	Result	
ACD fold, holo	monomer core Ca RMSD 1.54-2.44 A	pass
ACD fold, apo	3.36 A	marginal
Held-out peptide placement	**1.27 A**, centroid off by 0.77 A	pass
Groove overlap	**all 14 contacts among the 20 experimental groove residues**	pass
Contrast	held-out positive ipTM 0.899 vs best homopolymer negative 0.746	pass

One real defect: the peptide is threaded **backwards**. Forward-order RMSD is 16.49 A, reverse-order 1.27 A, and the per-residue deviations (25.5, 19.7, 13.3, 7.5, 1.2, 5.8, 12.7, 18.7, 25.0 A) are symmetric about the centre – the signature of an extended peptide lying in the right channel the wrong way round. Site, position and conformation are right; polarity is wrong. Neither quantity the pilot uses depends on direction, so the gate stands, but a model that cannot orient a peptide it has placed correctly is not a model to build a mechanism on.

A second symmetry trap had to be cleared first, the same one as in 3e: the ACD dimer has a two-fold axis, so its two grooves are equivalent by construction, and scoring against the wrong one reports the ~40 A distance between them as if it were an error.

Four predictions, fixed before the run finished; all four fail

	Prediction	Result
P1	interface score correlates positively with measured partition	**fails.** Every aggregation of every score lies between -0.17 and +0.02. The most reliable score reaches rho = -0.127 [-0.267, +0.021] – the wrong sign
P2	real windows beat their own composition-matched shuffles	**fails.** Real 0.612, shuffled 0.638, difference **-0.026** [-0.059, +0.006]; Wilcoxon p = 0.955 in the predicted direction
P3	structure adds over window composition	**fails.** Cross-validated rho: composition -0.059, structure -0.093, both -0.035

	Prediction	Result
P4	structure adds over the ESMC-6B + SAE sequence model	**fails.** Sequence model alone rho 0.335; adding structure *lowers* it to 0.232. Largest correlation of any structural score with the sequence model's residual is 0.134, wrong-signed, CI including zero

P2 is the one that matters most. If the co-fold cannot tell a real 16-mer from its own scramble, it is not reading sequence order, and a score that does not read sequence order cannot rank clients. For context, the largest residual correlation here (0.134) sits alongside 0.092 for AlphaFold monomer features (3d) and 0.087 for native-partner interfaces (3e) – three independent structural approaches landing in the same narrow band just above nothing.

The controls say what the score is actually measuring

Control	ipTM	Percentile among 4,577 real client windows
9ef3 IxI 9-mer (held out)	0.899	97.9
IxI 16-mer (held out)	0.790	77.9
poly-G 16-mer (negative)	**0.746**	**69.2**
6gjh N-terminal 5-mer (in training)	0.650	51.1
poly-S (negative)	0.614	44.7
6gjh N-terminal 16-mer (in training)	0.549	34.4
poly-EK (negative)	0.497	27.5
poly-K (negative)	0.299	5.4
poly-E (negative)	0.227	1.3

The evolved IxI motif does score at the top. But poly-glycine – a sequence with no side chains at all – outscores both in-training positives and 69 % of real client windows, while the negatives that do sit at the floor are the charged ones. On this evidence ipTM mostly separates "charged homopolymer" from "everything else", which is a far weaker axis than the gate alone suggested.

How much of the score is noise

Boltz-2 defaults to no seeding, so the pilot was one unseeded draw. Re-running 548 of the same windows is a second independent draw from the same model on the same input (`cm25`).

A window's ipTM moves by **0.140 on average between two runs of identical input**, against a spread across different windows of 0.185. Test-retest r = **0.497**: half the variance in the pilot's window scores is run-to-run noise.

That matters less than it sounds for a per-protein score built from ~22 windows, and more than it sounds for the aggregations that take extremes. Measured by split-half over each protein's own windows:

Per-protein score	Reliability	Ceiling on |rho|	Observed |rho|
groove_recall_top3	0.461	0.679	0.127
iptm_top3	0.359	0.599	0.047
groove_recall_mean	0.300	0.548	0.120
iptm_fhi	0.170	0.413	0.108
iptm_max	0.169	0.411	0.022
groove_recall_max	0.134	0.366	0.054
iptm_mean	0.115	0.340	0.135

Per-protein score	Reliability	Ceiling on |rho|	Observed |rho|
groove_recall_fhi	0.103	0.321	0.168

Three of the eight have ceilings of 0.55-0.68 and observe 0.05-0.13, so for those the null is real: the instrument had room to show an effect three to five times larger than it did. The other five are too noisy to interpret in either direction, and that includes `max`, which is the most biologically natural aggregation – "the best binding site on the protein" – because a maximum over noisy values is mostly a maximum over noise.

A selection trap worth recording. The score with the largest correlation, `groove_recall_fhi` at -0.168, is the single *least* reliable of the eight. Choosing a headline by effect size would have picked the noisiest feature; the figure selects on reliability instead, which does not depend on the outcome.

What this closes

The honest summary is a null with a caveat, not a clean null. Where the measurement is reliable enough to interpret, a predicted HSPB1-ACD / client-window interface carries no information about which proteins HSPB1 actually binds – it does not beat composition, does not beat a scramble, and subtracts from the sequence model. Where the measurement is not reliable enough, the tier is uninformative rather than negative.

Scaling Design A from 200 to 2,000 proteins would not fix this. The limit is not sample size; the limit is that the per-window score is half noise and does not respond to sequence order, and ten times more proteins leaves both properties unchanged. The failure of P2 in particular is a property of the model, not of the sample.

Two things could change the answer, and neither is a bigger version of this experiment. First, the ACD dimer alone omits the N-terminal region, which carries much of sHSP client binding and is disordered – the part a co-folding predictor is worst at, and the part no crystal structure constrains. Second, a co-folding model trained on native crystallised interfaces is being asked about a transient, low-affinity, non-native complex; the reversed threading of a peptide it otherwise placed to 1.27 A is a small visible symptom of that mismatch. Both point away from more structure prediction and back toward the data: the complex effect of 3c remains real, remains unexplained by occlusion (3e) or by a predicted chaperone interface (3f), and the next thing to separate is abundance from binding strength in the measurements themselves.

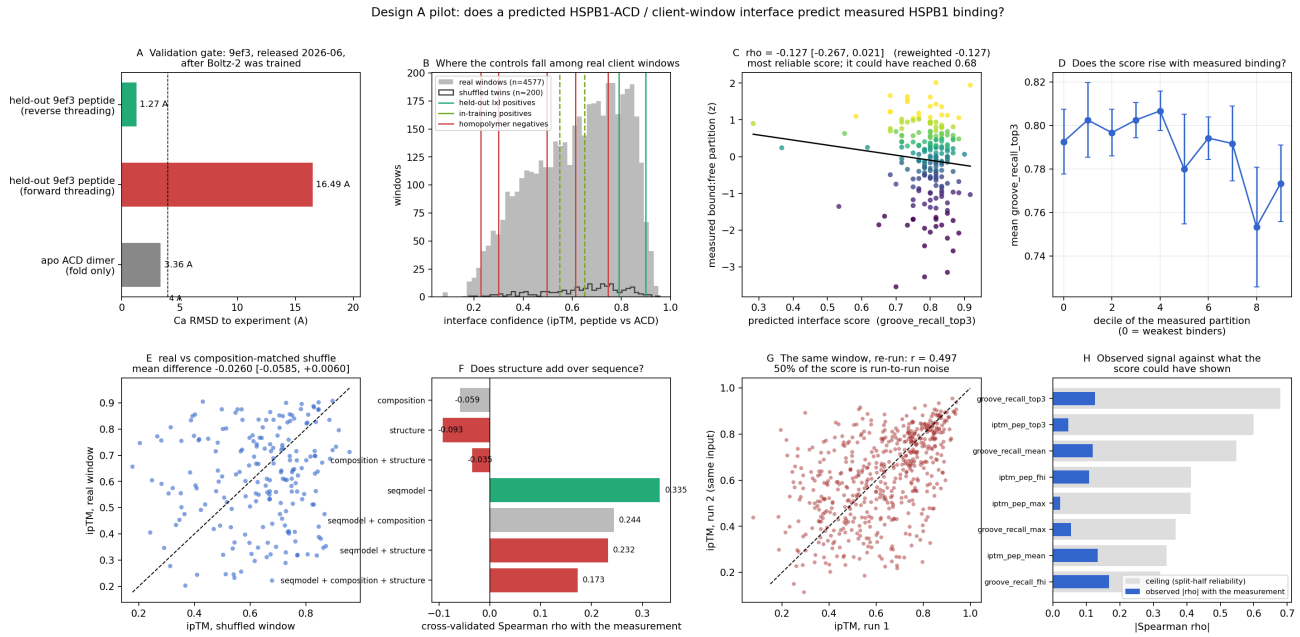


Figure 8: Design A pilot. A: the validation gate against 9ef3, released after Boltz-2 was trained. B: where every control falls among 4,577 real client windows. C, D: the most reliable per-protein score against the measured partition, as a scatter and by decile. E: real windows against their own composition-matched shuffles. F: cross-validated prediction from each feature block. G: the same window re-run, showing half the score is run-to-run noise. H: observed correlation against the ceiling each score's reliability allows.

4. What predicts the pull-down with no perturbation

A pull-down level mixes how much protein is in the tube (abundance) with how strongly it binds. Both parts were tested with named, readable properties, alone and with the sequence model (`cm07`).

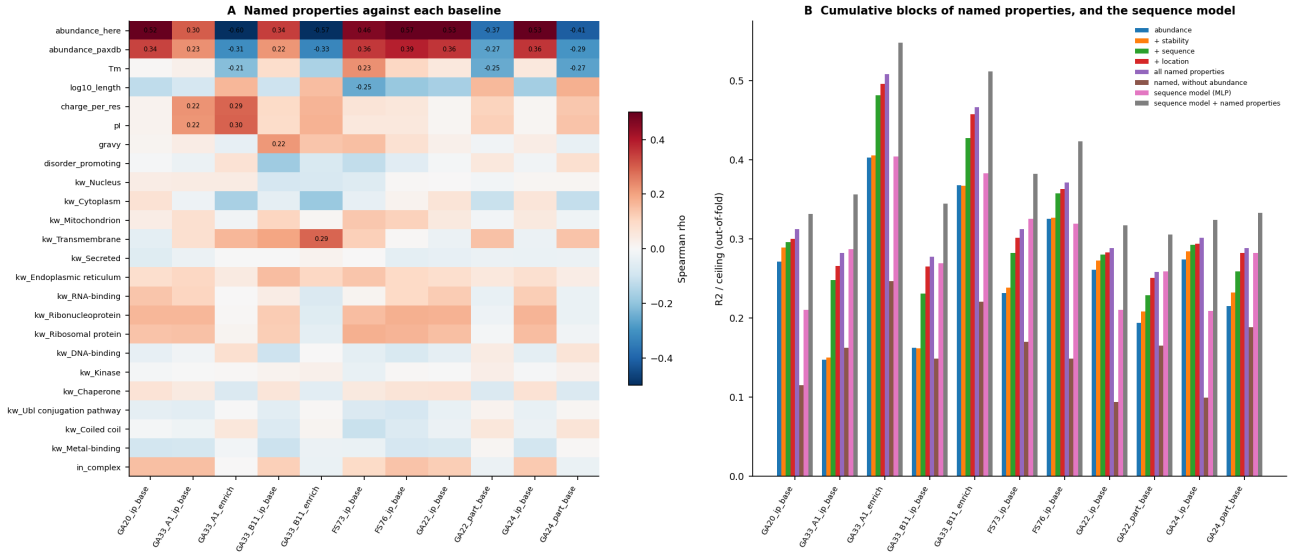


Figure 9: A: Spearman rho of each named property with each baseline. B: out-of-fold R^2 / ceiling as property blocks are added, the sequence MLP alone, and the two combined.

Predictor set	HSPB1 pull-down	DNAJA1/B11 pull-down	Binding over abundance
Abundance (here + PaxDb)	0.23-0.32	0.15-0.16	0.19-0.40
+ Tm, sequence physics, location	0.28-0.36	0.26	0.25-0.49
All named properties	0.29-0.37	0.27-0.28	0.26-0.50
Named properties without abundance	0.09-0.17	0.15-0.16	0.16-0.25
Sequence MLP alone	0.21-0.32	0.27-0.28	0.26-0.40
Sequence MLP + named properties	**0.31-0.42**	**0.34-0.35**	**0.30-0.54**

Values are out-of-fold R^2. The ceilings here are 0.98-0.998, so R^2 / ceiling differs by under 0.01.

What the properties show:

- **Abundance dominates the raw pull-down.** It correlates with every HSPB1 baseline (rho 0.46-0.57).
- **It reverses for enrichment.** Abundant proteins are less enriched over input (rho −0.57 to −0.60 in GA_33). Capture rises less than proportionally with amount, as found before (slope about 0.35).
- **Abundance-free binding** rises with transmembrane segments and with positive charge (DNAJA1: rho +0.30 with pI), and falls with stability (Tm, rho −0.21 to −0.27).
- **Ribonucleoprotein and complex membership** raise the raw HSPB1 pull-down slightly.

Sequence predicts the abundance-free binding about as well as all the named properties together, and the two combined are best. So sequence carries most of what the named properties do, and somewhat more.

5. One representation across conditions

5.1 Leave-one-run-out transfer

Six trunks were each trained without one run's targets. That run's targets were then read out from the frozen latent by ridge, within each fold, since fold models' latents are not aligned with each other.

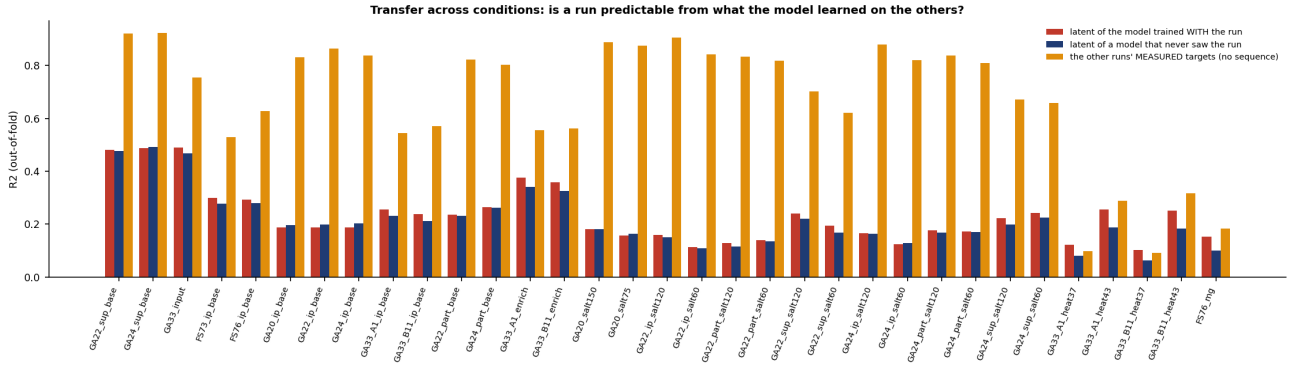


Figure 10: For each target: read-out from the latent of the model trained with the run (red), from a trunk that never saw the run (blue), and, for scale, the prediction from the other runs' MEASURED targets with no sequence (orange).

Run left out	Its targets, read from a trunk that never saw it (with the run)
GA_20 (HSPB1, salt)	baseline 0.20 (0.19); salt 75 / 150: 0.16 / 0.18 (0.16 / 0.18)
GA_22, GA_24 (HSPB1, salt; with supernatant and partition)	within 0.00-0.03 of trained-with for all 18 targets
FS73 (HSPB1)	baseline 0.28 (0.30)
FS76 (HSPB1, Mg)	baseline 0.28 (0.29); **Mg 0.10 (0.15)**
GA_33 (DNAJA1/B11, heat)	abundance 0.47 (0.49); enrichment 0.32-0.34 (0.36-0.38); **heat 43 °C 0.18-0.19 (0.25-0.26)**

One shared representation covers every HSPB1 baseline and salt condition: no run teaches it something the others do not. What it loses when a run is left out is a condition type found nowhere else. Heat exists only in GA_33 and Mg only in FS76. The two DNAJ baits also lose a little on their baselines.

The measured cross-run prediction (orange) shows how much the conditions share as data:

- salt responses are 70-93% predictable from the other salt experiments' measurements;
- heat is only 9-34% predictable from anything else measured.

Heat is the condition least like the others.

5.2 Which conditions the model represents along the same axes

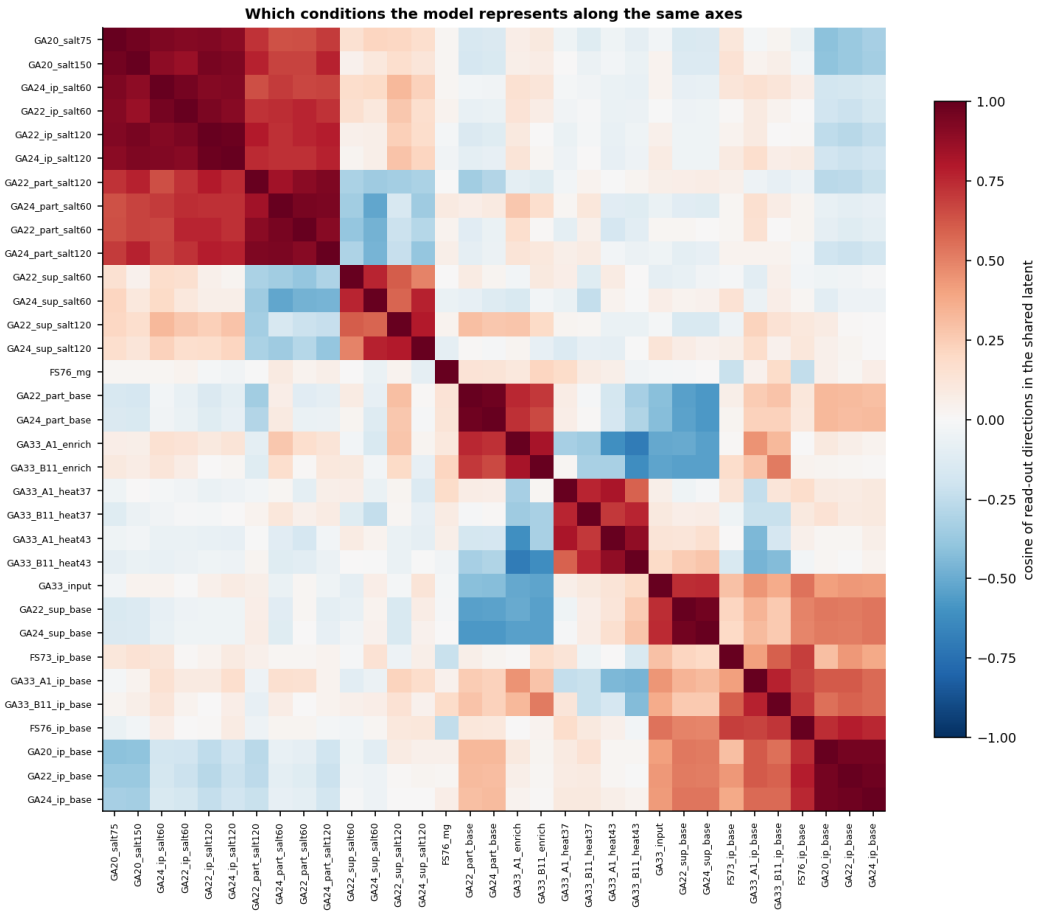


Figure 11: Cosine between targets' read-out directions in the shared latent, clustered.

The latent organises the conditions into a few axes:

- **Salt, pull-down and partition:** one axis shared by all three salt experiments (GA_20, GA_22, GA_24).
- **Salt, supernatant:** a separate axis. Salt changes the free pool along different directions from binding, as the GA_22/24 report found from the data.
- **Abundance and baselines:** an abundance axis carrying the supernatant and input levels, with the baselines close to it.
- **Enrichment and partition:** pointing opposite to abundance.
- **Heat:** an axis of its own.
- **Mg:** aligned with nothing.

5.3 Regressing the baseline out

Two versions, which agree.

(a) In the model. Within each fold, the subspace of the latent that predicts abundance, baselines and enrichment (rank 14) was projected out, and each effect read again. Effects barely change:

- GA_20 salt: 0.16 → 0.15;

- heat 43 °C: 0.25 → 0.24;
- Mg: 0.15 → 0.13.

The effect signal sits in directions separate from the baseline signal.

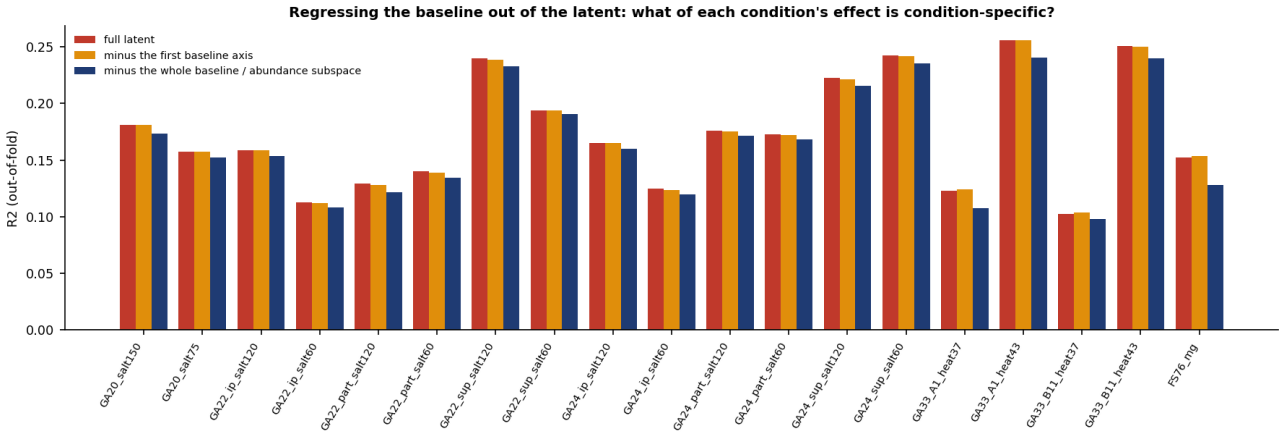


Figure 12: Effect targets read from the full latent, from the latent minus its first baseline axis, and minus the whole baseline / abundance subspace.

(b) In the data. Each effect was residualised on its own run's measured baseline and abundance, fitted on training proteins only, and a fresh MLP was trained on the residuals.

Effect and baseline in one run share their control samples, and therefore their measurement noise. So I repeated it with baselines from an independent experiment: GA_24's for GA_22, FS73's for GA_20 and FS76, the other bait's for GA_33.

Effect	Variance left after removing baseline (own run / independent run)	Sequence R^2, raw → residual (own / independent)
Heat 43 °C, DNAJA1	0.52 / 0.68	0.27 → 0.13 / 0.20
Heat 43 °C, DNAJB11	0.52 / 0.52	0.26 → 0.13 / 0.10
Heat 37 °C	0.86-0.88 / 0.85-0.97	0.11-0.13 → 0.07-0.11
Salt, pull-down (GA_20, GA_22, GA_24)	0.83-0.94 / 0.90-0.94	essentially unchanged
Salt, supernatant	0.83-0.99 / 0.84-0.99	unchanged or higher (0.25 → 0.28-0.31)
Mg	0.92 / 0.97	0.17 → 0.16 / 0.15

The heat response at 43 °C is about half a starting-level effect. Proteins' 43 °C change depends on their 35 °C level, even when that level comes from another experiment. About half of what sequence predicts about heat goes through that level.

Salt and Mg responses are condition-specific. Almost none of their variance is explained by the baseline, and what sequence predicts about them is untouched by removing it.

6. What the SAE features say

Gradient attribution through an SAE-input model was unstable from fold to fold (r 0.19-0.29). The 16,384 features are highly redundant, and different fold models credit different members of correlated groups. So the

features are ranked by a stable statistic instead: Spearman rho with the family-averaged target. Its agreement between two halves of the proteins split by sequence cluster:

Family	Halves r	Top-50 overlap
Abundance	0.93	0.82
Pull-down baseline	0.84	0.78
Bound:free and enrichment	0.90	0.56
Salt	0.73	0.72
Mg	0.79	0.68
Temperature	0.87	0.24

Labels are Biohub's LLM-generated annotations, which are hypotheses. Each is checked against the UniProt keywords of the proteins on which the feature is active.

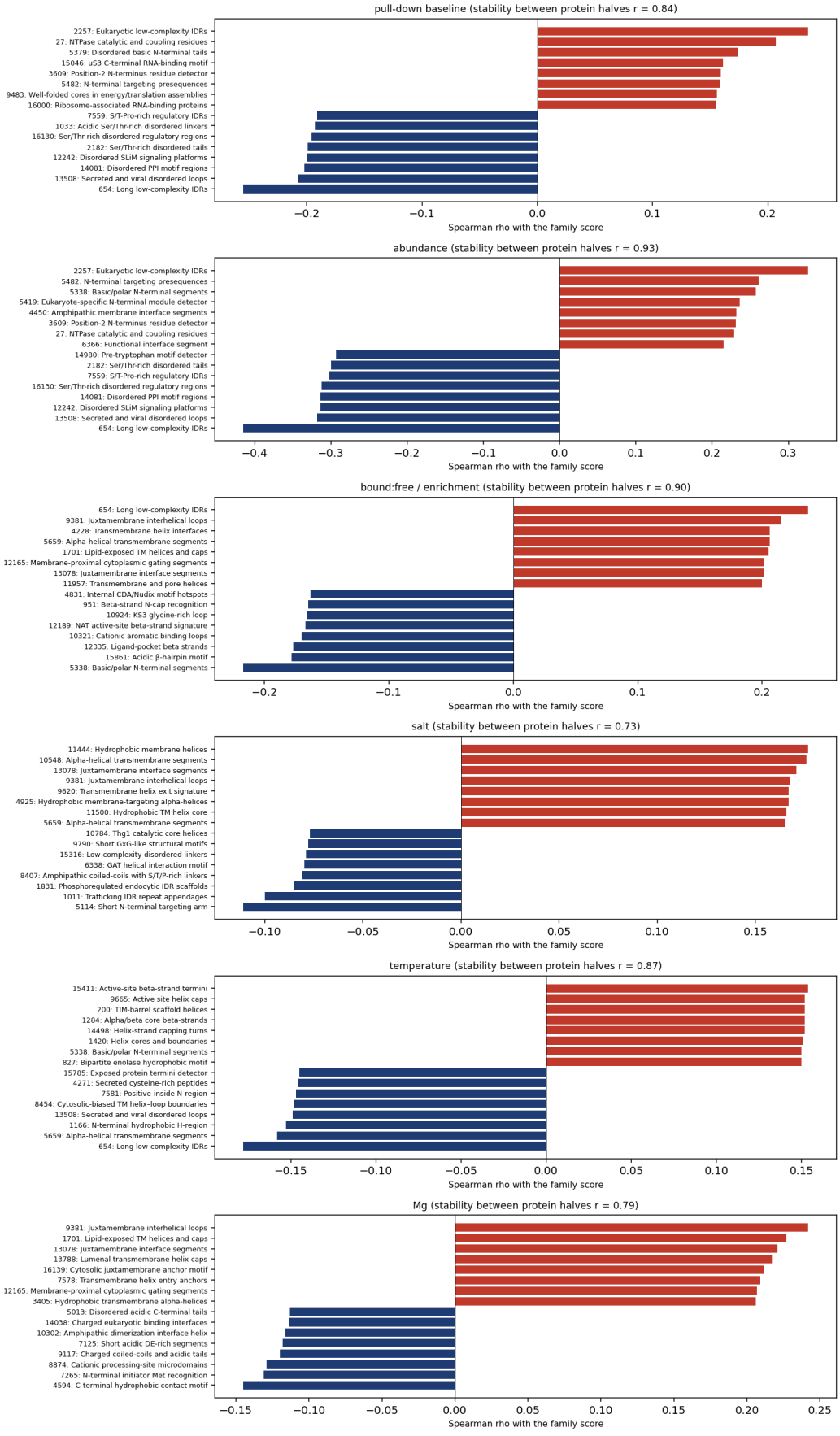


Figure 13: The 8 features most positively and most negatively associated with each family, with their Biohub labels.

Family	Higher with (label; keyword check)	Lower with
Pull-down baseline, abundance	low-complexity IDRs typical of eukaryotic enzymes; NTPase catalytic residues (nucleotide-binding); basic N-terminal tails and ribosomal RNA-binding motifs (ribosomal proteins)	Ser/Thr-rich, S/T-Pro-rich and SLiM disordered regions (59-78% phosphoproteins); secreted disordered loops
Bound:free / enrichment	transmembrane helices, juxtamembrane loops (62-74% membrane proteins); long low-complexity IDRs	enzyme ligand-pocket β-strands and active-site loops (hydrolases); basic N-terminal mitochondrial segments
Salt	hydrophobic and juxtamembrane transmembrane helices (72-82% membrane)	phosphoregulated trafficking / endocytic IDR scaffolds
Temperature (43 °C)	α/β enzyme cores, TIM barrels, active-site helix caps (nucleotide-binding, hydrolases)	membrane, secreted, signal-peptide and disordered features
Mg	transmembrane-helix features (61-75% membrane)	acidic tails, charged coiled coils, nuclear interfaces

Two cautions.

- The membrane axis appears in salt, Mg and enrichment alike. It may describe how membrane proteins behave in a detergent-free lysate (aggregation, insolubility) as much as chaperone binding.
- The heat features (structured α/β enzyme cores) fit thermal unfolding, but the temperature ranking's top features are the least stable of any family (top-50 overlap 0.24). Read them as a class, not as individual features.

7. A proteome-wide prediction table

The five fold models were applied to the 8,836 reviewed human proteins that no run measured; each fold model was trained on 80% of the measured proteins. This adds predictions for every target to the out-of-fold predictions for the 10,777 measured proteins: 19,613 proteins in `data/campaign/proteome_predictions.tsv`.

The expected accuracy is each target's out-of-fold R^2 (0.10-0.49). The table is a prior for picking candidates, not a substitute for measurement.

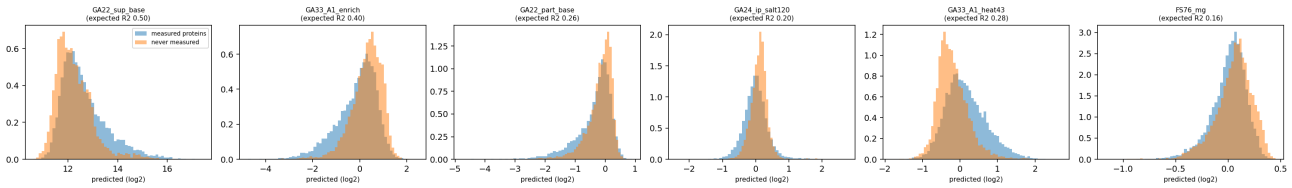


Figure 14: Predicted distributions for measured and never-measured proteins, for six targets.

8. Critical evaluation

1. **One protein-level representation.** ESMC-6B mean-pooled states carry about a quarter of the reliable variance, and nothing tried here moved that. The rest is either not in the sequence (lysate state, complex membership, post-translational state, local concentration) or not in this representation. The measured covariates show the first is substantial for baselines.
2. **Abundance as input or target.** For baselines, measured abundance is the strongest predictor, and sequence partly predicts it. The fair question is what sequence adds beyond abundance. That is small for baselines (+0.012, interval includes 0) and real for salt and Mg.

3. **Shared measurement noise.** In-run residualisation shares control samples between effect and baseline. The independent-baseline version removes that and gives the same answer.
4. **Not every run measures abundance.** HSPB1's FS73 / FS76 and GA_20 have no supernatant or input, so abundance comes from GA_33's lysate and PaxDb. Different lysate preparations add noise.
5. **The SAE is not an accurate input.** It is the most interpretable one, but on its own it predicts less (0.18) than the dense layers. Its feature associations describe the measurements, not the model's predictions.
6. **Bugs found and fixed during the campaign:**
   - a shared row-index file overwritten by later specs, which zeroed the first latent read-out;
   - a drift-covariate aliasing in the linear model on sparse proteins (from the GA_22/24 work);
   - variable shadowing in the per-residue writer, caught by its own completeness assertion.

   Each was caught by a check before any number was reported.

9. Files

- **Code:**
  - `cm00_union.py`, `cm01_embed.py`, `cm01b_residues.py` (embeddings, SAE, per-residue states);
  - `cm02_targets.py`, `cm02b_features.py` (targets, features, folds);
  - `cm03_mlp.py` (population MLP, covariates, residualisation), `cm03b_ridge.py`;
  - `cm04_latent.py` (transfer, regress-out, geometry, measured cross-run);
  - `cm05_attrib.py`, `cm05b_assoc.py`, `cm05c_annot.py` (SAE);
  - `cm06_attn.py` (attention pooling), `cm07_baseline.py` (named properties);
  - `cm08_figures.py`, `cm09_stats.py` (bootstrap), `cm10_proteome.py`;
  - `cm12_complex.py`, `cm16_network.py` (complex and network context, 3c);
  - `cm13_structure.py`, `cm15_maskpool.py`, `cm14_struct_test.py` (AlphaFold structure, 3d);
  - `cm17_pdb_index.py`, `cm18_interface.py`, `cm19_iface_test.py` (experimental interfaces, 3e);
  - `cm20_pilot_select.py`, `cm21_windows.py` (pilot set and co-folding jobs, 3f), `cm22_val_score.py` (validation gate against 9ef3), `cm23_pilot_test.py` (the four predictions), `cm24_pilot_figure.py`, `cm25_reliability.py` (test-retest and split-half ceilings);
  - `cm_launch.py` and `cm_summary.py` (sweeps);
  - `slurm/cm0*.sbatch`, `slurm/cm1[3689].sbatch`, `slurm/boltz_val.sbatch`, `slurm/boltz_pilot.sbatch`, `slurm/boltz_rep.sbatch`.
- **Results:** `data/campaign/`:
  - `runs/` (every group: config, R^2, out-of-fold predictions, latents, saved models);
  - `ridge.json`, `cm07_baseline.json`, `cm09_stats.json`;
  - `attrib/` (associations, annotated features);
  - `cm12_complex.json`, `cm16_network.json`, `cm14_struct.json`, `cm19_iface.json`;
  - `cm20_pilot.json`, `cm22_val.tsv`, `cm23_pilot.json`, `cm25_reliability.json`;
  - `pilot_proteins.tsv`, `boltz_jobs.tsv`, `pilot_windows.tsv` (4,787 co-folds), `pilot_scores.tsv`, `rep_pairs.tsv`;
  - `boltz/` (4,787 + 600 predicted structures and confidence files, 2.1 GB);
  - `struct.tsv`, `interface.tsv`, `iface_entry.tsv`, `pdb_manifest.tsv`;
  - `proteome_predictions.tsv`.
- **External data:** `data/external/`: `complexes/` (Complex Portal), `string/`, `sifts/`, `afdb_human/` (AlphaFold human proteome tar), `pdb_assemblies/cif/` (29,816 RCSB biological assemblies, 11 GB, fetched by `code/fetch_pdb_assemblies.sh`). Boltz-2 weights live in `~/.boltz` (3.7 GB), and the co-folding environment is the `boltz` conda env (boltz 2.2.1, torch 2.14.1+cu130, cuequivariance kernels; CUDA 13 is required by the Blackwell RTX PRO 6000 cards).
- **Figures:** `reports/figures/campaign/`.